INCH-POUND
MIL-DTL-44104B
14 December 2022
SUPERSEDING
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20 October 1987

DETAIL SPECIFICATION

MAT, SLEEPING

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers one (1) type and size sleeping mat.

1.2 Classification. Not applicable.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50199 - Thread, Polyester Core, Cotton or Polyester-Covered

Comments, suggestions, or questions on this document should be addressed: Attn: DLA Troop Support, 700 Robbins Street, Philadelphia, PA 19111-5096. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

AMSC N/A FSC 8465

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5038 - Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-PRF-20696 - Cloth, Waterproof, Weather Resistant
MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage (General Use)

(Copies of these documents are available online at https://quicksearch.dla.mil)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY, DEVCOM - SOLDIER CENTER

2-3-438 - Mat, Sleeping

(Copies of drawings are available from the U.S. Army, DEVCOM Soldier Center, ATTN: FCDD-SCP-PEI, 10 General Greene Avenue, Natick, MA 01760-5019.)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org)

ASTM INTERNATIONAL

ASTM D3575 - Standard Test Methods for Flexible Cellular Materials Made from Olefin Polymers
ASTM D3775 - Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M - Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D6193 - Standard Practice for Stitches and Seams

(Copies of these documents are available online at https://www.astm.org)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Unless a specific exemption has been obtained, nothing in this document, supersedes applicable laws and regulations.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.1.

3.2 Conformance inspection. When specified, a sample shall be subjected to inspection in accordance with 4.3.2.

3.3 Standard sample. Unless otherwise indicated, the finished mat shall match the standard sample for shade and appearance, and shall, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.5 Design. The sleeping mat shall conform in all respects to Drawing 2-3-438 and as specified herein.

3.6 Material(s).

3.6.1 Base nylon material (coated and uncoated). The nylon cloth shall meet the requirements in Table I when tested as specified in 4.5.

TABLE I. Physical requirements of base nylon material.

Characteristic	Requirement
Yarns per inch: Warp Filling	 28 -33 28 – 33
Weight, oz./sq.yd.: Uncoated Coated	 5.9 – 10.0 15.0 – 19.2

3.6.1.1 Alternate base nylon cloth (coated and uncoated). As an alternate to the material specified in 3.6.1, the material for the patches shall conform to Type II, Class 1 or 3 of MIL-PRF-20696, except that the base fabric shall be nylon.

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3.6.2 Secondary material(s) and components.

3.6.2.1 Body material. The body of the mat shall be closed cell expanded cross-linked polyethylene foam. The material shall be physically blown using nitrogen, or a chemically blown commercial application which is non-toxic, and whose products of decomposition are non-toxic. The physical properties of the body of the mat shall meet the requirements in Table II when tested as specified in 4.5. The length of the mat shall not contain more than one (1) lateral weld.

TABLE II. Physical requirements of body material.

Characteristic	Requirement
Tensile strength, psi., (min.)	70
Elongation, percent, (min.)	100
Tearing resistance, lbs./inch, (min.)	15
Density, lbs./cu.ft., (max.)	3.6
Heat shrinkage, percent, (max.)	15
Compression strain: 10 psi., percent, (max.) 30 psi., percent, (max.)	 35 65
Compression set: 22 hrs. at room temperature, percent, (max.) 22 hrs. at 0°F, percent, (max.)	 45 55
Water absorption, percent, (max.)	17
Weld (breaking) strength per inch width, (min.)	16.9

3.6.3 Patches. The patches shall be cut from a polychloroprene coated woven nylon cloth which has been uniformly coated on both sides.

3.6.4 Adhesive. The patch adhesive shall be a polychloroprene based self-vulcanizing adhesive, or an approved commercial alternative. The patch adhesion (peel force required to remove patches from the mat) shall not be less than 11.2 pounds when tested as specified in 4.7.4.

3.6.5 Tape. The tape used to secure the patches shall be 3/4-inch wide, conforming to MIL-DTL-5038.

3.6.6 Thread. The thread shall conform to Type I or II, Tex 106-150, 2, 3 or 4 plies of A-A-50199.

3.7 Color. The color shall be Olive Drab, Foliage Green 504 or as specified in the contract or procurement documents. The finished cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced.

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3.7.1 Visual shade matching. The color and appearance shall match the standard sample (see 6.5) when tested as specified in 4.7.1.

3.8 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished mat shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.8.

3.9 Label. The sleeping mat shall have a marking as specified on Drawing 2-3-438.

3.9.1 Barcode hang tag. Each mat shall have a barcode label/hangtag conforming to Type VII or VIII, Class 17 of MIL-DTL-32075. The barcoding element shall be a 13-digit National Stock Number (NSN). There shall be a 12-digit Universal Product Code (UPC) number assigned for all NSNs by the contracting activity. The initials "UPC" must appear beneath the code. The barcode for NSN and UPC type shall be a medium to high density and shall be located so that it is completely visible on the mat when it is folded or packaged as specified and so it causes no damage to the mat.

310 Construction.

310.1 Stitches, seams and stitching. All stitches, seams and stitching shall conform to Type 301 of ASTM D6193. All seams shall have 6 to 8 stitches per inch. The type of seam and stitching shall conform to Drawing 2-3-438 and as specified herein. End of stitching shall be overlapped 1/2-inch to prevent raveling.

3.10.2 Thread tension. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.10.3 Repairs of type 301 stitching. Repairs of Type 301 stitching shall be as follows:

a. When thread breaks, skipped stitches, run-offs, or bobbin runouts occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2-inch back of the end of the stitching. 1/

b. Thread breaks or two (2) or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2-inch in back of the defective areas, continue over the defective area and continue a minimum of 1/2-inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and re-stitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.10.3.1 Thread ends. All thread ends shall be trimmed to 1/4-inch maximum length.

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3.11 Securing tapes. The position and length of the tapes shall conform to Drawing 2-3-438. Ends of tape shall be seared.

3.12 Adhesive patches. The patches shall be cut from the coated nylon cloth (see 3.6.1) and secured to the mat with adhesive (see 3.6.4) by abrading the surface of the mat. The location and dimensions of the patches shall conform to Drawing 2-3-438.

3.13 Workmanship. The finished mat shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects, as specified in 4.4, shall not exceed the criteria specified in the contract or procurement documents.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1)
b. Conformance inspection (see 4.3.2)

4.2 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.3 Types of inspections.

4.3.1 First article inspection. The first article, submitted in accordance with 3.1 shall be inspected for appearance, color, finished measurements as specified in Drawing 2-3-438, examination for defects in 4.4 and tested for the characteristics as specified in 4.5 through 4.8.

4.3.2 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5 through 4.8 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits as specified in the contract and/or procurement documents, except where otherwise indicated (see 6.2).

4.4 Visual examination. The finished mat shall be examined for the defects listed in Table III.

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TABLE III. Defects.

Examination	Defect	Classification	
		Major	Minor
Materials			
	- Incorrect, not as specified	101	
	- Any cut, split, tear, hole, or crack, not uniform, contains large voids	102	
	- More than one lateral weld in any mat, weld not complete	103	
	- Nylon cloth not coated	104	
	- Color not as specified	105	
Components			
Thread	- Not as specified		201
	- Color not as specified	106	
Securing tapes	- Not as specified	107	
	- Any tape missing or not properly located	108	
	- Ends not seared		202
	- Color not as specified	109	
Adhesive patches	- Any patch missing, incorrect size, or not properly located	110	
	- Not adhered over entire area	111	
	- Not uniformly coated on both sides		203
	- Color not as specified	112	
Construction			
Mat design	-Not as specified, not in accordance with Drawing 2-3-438	113	
Seams and stitching	- Open seams, puckered, distorted, wavy, twisted, or irregular	114	
	- Loose or tight stitch tension		204
	- Missing stitches greater than ¼-inch length	115	
	- Stitching not as specified	116	
	- Stitching seams incorrectly repaired		205
	- Number of stitches per inch more or less than required	117	
	- Not overstitched as required		206
Cleanness and workmanship	- Spot/stain affecting appearance or serviceability	118	
	-Visible thread ends not trimmed to a length of not less than ¼-inch	119	
	- Objectionable odor		207
Packaging	- Not packaged in accordance with contract or procurement documents		208

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4.5 Component and end item testing. Testing shall be in accordance with the characteristics listed in Table IV. The methods of testing as specified wherever applicable and as listed in Table IV shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be one (1) continuous yard full width of the finished material for all physical and chemical tests. The sample size shall be as specified below (component only). For end item testing, the lot size shall be expressed in units of one (1) finished mat. The sample unit shall be one (1) mat. Any lot shall be unacceptable if one or more sample units (component and end item) or the lot average fail to meet any requirement specified.

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. Component and end item tests.

Characteristic	Requirement reference	Test method
Component		
Base nylon material:		
Yarns per inch	3.6.1	ASTM D3775
Weight: Uncoated Coated	 3.6.1 3.6.1	 ASTM D3776/D3776M, Option A ASTM D3776/D3776M, Option C
Coating	3.6.1	Visual (see 6.4)
Body material (end item)		
Tensile strength	3.6.2.1	ASTM D3575
Elongation	3.6.2.1	ASTM D3575
Tearing resistance	3.6.2.1	ASTM D3575
Density	3.6.2.1	ASTM D3575
Heat shrinkage	3.6.2.1	See 4.7.2
Compression strain: 10 psi 30 psi	3.6.2.1	ASTM D3575
Water absorption	3.6.2.1	ASTM D3575
Weld strength	3.6.2.1	See 4.7.3
Patch adhesive	3.6.4	See 4.7.4
Toxicity	3.8	See 4.8
Construction		
Seam type and stitching	3.10.1	Drawing 2-3-238

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4.6 End item dimensional examination. The finished mat shall be examined for conformance to the dimensions as specified on Drawing 2-3-438. Any dimensions not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of mats. The sample unit shall be one (1) finished mat.

4.7 Methods of inspection.

4.7.1 Visual shade matching. The color and appearance of the mat shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.5) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the mat shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 6.5.1). The mat shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

4.7.2 Heat shrinkage. The specimen shall be tested according to the method specified in 4.7.2.1 through 4.7.2.4 for heat shrinkage is as follows:

4.7.2.1 Apparatus.

a. 12-inch ruler graduated in 0.1-inch dimension.
b. Air circulating oven capable of maintaining 200 (± 2) ºC.
c. Aluminum or steel plate about 14-inches square and approximately 0.060-inch width.

4.7.2.2 Specimen preparation. Three (3) test specimens, each measuring approximately 10-inches by 1-inch by 3/8-inch shall be cut using a die stamp or other suitable means from the sample. Samples shall not be tested less than 24 hours after manufacture. The specimen shall be marked by any suitable means about 1-inch from each end of the specimen and the distance between the two marks shall be measured to the nearest 0.1-inch.

4.7.2.3 Procedure. The three (3) specimens shall be lightly dusted with French chalk and placed in an oven at 200 (± 2) ºC on a preheated aluminum or steel plate similarly dusted with French chalk. After 5 minutes (± 2) seconds the plate and specimens shall be removed from the oven and the specimens tipped, as quickly as possible, onto a non-conducting surface such as asbestos and allowed to cool for 15 minutes. The metal plate shall then be re-dusted and replaced in the oven. No further tests shall be performed until both the oven and metal plate have remained at 200 ºC for at least 5 minutes. The timing of the heating period shall commence when the oven is closed and finished when the door is opened. Any temperature drop due to the insertion of the test specimens shall be ignored. No more than six (6) specimens shall be tested at any one time. After the specimens have been allowed to cool for at least 15 minutes, the distance between the marks shall be measured to the nearest 0.1-inch.

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4.7.2.4 Calculation of results. The heat shrinkage shall be calculated as follows:

$$S = \frac{L_1 - L_2}{L_2} \times 100$$

S = Percentage heat shrinkage
L1 = Original length in inches
L2 = Final length in inches

4.7.3 Weld strength. Specimens shall be cut, 1-inch wide by 6-inches long with the weld perpendicular to and centered along the 6-inch direction and tested in a constant strain tensile machine at a rate of separation of the jaws of 2-inches per minute. The weld strength shall be reported to the nearest 1-pound (breaking strength) per inch.

4.7.4 Patch adhesion. Adhesion of the patches to the body of the mat shall be checked by cutting specimens, at the point of attachment of the patches to the body of the mat, equal in width to the superimposed patches, and allowing a free length of mat material to remain attached at least 6-inches in length. Tapes should similarly be left attached. Specimen shall be modified at the trailing edge of the patch by severing completely through the nylon tape and mat body material, leaving the patch bonded to the mat body with remainder of the nylon tape and mat body material pointing normally in the same direction. The free end of the tape shall be mounted in the top jaw of a constant strain tensile machine. The free end of the mat body shall be bent trough 180 degrees and inserted in the bottom jaw of the machine. Local trimming of the mat body material may be necessary to facilitate fitting the mat body material into the jaw. The machine shall be set to run at a rate of separation of the jaws of 2-inches per minute and the peel strength determined. The results shall be reported to the nearest 0.1 pound. Adhesion shall be determined not less than seven (7) days after manufacture.

4.8 Toxicity test. When the toxicity requirement (see 3.8) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.4 and 6.6).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The sleeping mat is intended for use as a ground cover in conjunction with the sleeping bag.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced (see 2.1)
c. When first article is required (see 3.1)
d. When Toxicity is required (see 3.8, 6.4 and 6.6)
e. Inspection conditions (see 4.2)
f. Performance inspection with acceptance quality limits (AQL) (see 4.3.2)
g. Packaging (see 5.1)

6.3 Standard sample. For access to samples and drawing, address the contracting activity issuing the invitation for bids or request for proposal.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for that specified requirement.

6.5 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be “The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens”.

6.5.1 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75.

6.6 Toxicity testing. If dermal toxicity testing is required, it will be conducted using Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.or/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects 20745788?page=1)

Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100)
Dermal Toxicity (https://www.law.cornell.edu/cfr/text/40/798.2250)

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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6.8 Subject term (key word) listing.

Coating
Foam
Nylon

Custodians:
Army – GL
Navy – NU
Air Force – 11

Preparing activity:
DLA – CT

(Project 8465-2022-101)

Review activities:
Army – MD
Navy – MC

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